

Review for Exam #1

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MATH 241

Lecture #8 — Ch. 1-2

List of Topics

- ① (A.1): Set notation, including the use of unions, intersections, and complements
- ① (Ch. 1: 1.1-1.4, 1.6): Probability and counting
 - ▶ Sample spaces, outcomes, and events
 - ▶ Constructing new events from the unions, intersections, and complements of other events
 - ▶ Probability functions and the axioms of probability
 - ▶ Additional properties: complements, generalized additive property, and inclusion-exclusion
 - ▶ Combinations, permutations, and the multiplication principle
- ② (Ch. 2: 2.1-2.6): Conditional probability
 - ▶ Definition and its properties
 - ▶ Use of Law of Total Property, Multiplication Rule, and Bayes' Theorem
 - ▶ Independence

Practice Problem

From Grimmett & Welsh: You are traveling on a train with your sister. Neither of you has a valid ticket and the inspector has caught you both. A special punishment is given: each of you two will choose from a box initially containing 9 chocolates, 3 of which are contaminated with a deadly poison.

- 1 If you choose first (before your sister), what is the probability that you will survive?
- 2 If you choose first and survive, what is the probability that your sister survives?
- 3 If you choose first and die, what is the probability that your sister survives?
- 4 Is it in your best interest to persuade your sister to choose first (assuming you want to live)?
- 5 If you choose first, what is the probability that you survive, given that your sister survives?

Practice Problem #2

From Grimmett & Welsh: Which of the following is more probable:

- Getting at least one six with 4 throws of a die, or
- Getting at least one double six with 24 throws of two dice?

Practice Problem #3

From Grimmett & Welsh: A single card is removed at random from a deck of 52 cards. From the remainder, we draw two cards at random and find that they are both spades. What is the probability that the first card removed was also a spade?

Practice Problem #4

From Grimmett & Welsh: A rare, but potentially fatal disease has an incidence of 1 in 10^5 in the population at large. There is a diagnostic test however. If you have the disease, the test is positive 90% of the time. If you do not, the test is positive 5% of the time. Your test result is positive. What is the probability that you have the disease?

For Rest of Period

- Work on review problems from study guide & homework #3
- Ask any questions regarding material or problems
- Office hours on:
 - ▶ Friday from 10:30-11:30am
 - ▶ Monday from 1-2:30pm